

host_agent

Here are the results for commercial buildings in Exeter that are taller than 10 meters:

- 200 commercial buildings in Exeter meet the criteria (relative roof base height > 10m).
- The map displays all these buildings, each with interactive details such as building ID, use, and height.

Would you like to view or download the interactive map showing these buildings? If you need further analysis or details on specific buildings, let me know!

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are commercial buildings, and the condition is that they must be taller than 10m.

output steps: ["Find Exeter 1 area", "Find commercial buildings in Exeter as many search results", "Apply condition for buildings taller than 10m"]

named_area_agent

I found the area polygon for Exeter. The filtered results are saved in the artifact named `filtered_exeter_area_polygon`.

data_analysis_agent

Filtered Exeter area polygons: 34 entries found. - Artifact name: `filtered_exeter_area_polygon` - Description: Polygon(s) from `exeter_area_polygon` specifically representing Exeter, filtered using `name1_text`, `description`, and `descriptiongroup`.

Filtered commercial buildings in Exeter with relative roof base height > 10m. 200 out of 1814 buildings matched. - Artifact name: `commercial_buildings_exeter_over_10m` - Description: GeoDataFrame of commercial buildings in Exeter where the relative roof base height ('`height_relativeroofbase_m`') is greater than 10 meters. Includes all original columns and geometry.

buildings_agent

Search Results: - 72,895 commercial buildings were found in the area defined by '`filtered_exeter_area_polygon`'. - After filtering for buildings with a relative roof base height greater than 10 meters, 200 commercial buildings matched your criteria.

Filtered Artifact: - Artifact name: commercial_buildings_exeter_over_10m - Description: GeoDataFrame of commercial buildings in Exeter where the relative roof base height ('height_relativeroofbase_m') is greater than 10 meters. Includes all original columns and geometry.

Let me know if you need further analysis or details on these buildings!

plotting_agent

Plotted 200 commercial buildings in Exeter over 10m in height. First 5 entries:

osid	buildinguse	height_relativeroofbase_m	height_absoluteroofbase_m
af861363-b5de-480c-83ba-946c2a7b97a3	Commercial Activity: Retail	10.2	12.6
03e10934-42f4-43c7-93bd-ab90e3444303	Commercial Activity: Distribution Or Storage	13.2	20.0
79b6513f-b2d3-44a7-b178-8eee64207b24	Commercial Activity: Other	10.8	59.9
d3358ccb-a2d3-4e56-ae67-a59ee4bb06d3	Commercial Activity: Other	11.3	61.4
062fc15b-c91b-4245-9ac4-339e660ffb2e	Commercial Activity: Other	11.1	47.7

'commercial_buildings_exeter_over_10m_map.html'

Folium map showing all commercial buildings in Exeter with a relative roof base height greater than 10m. Each building is interactive with tooltip showing ID, use, and height.

Tool Calls and Code

Tool: send_message

- Args: {'target': 'planning_agent', 'task_description': 'Provide the general steps to solve the query: show commercial buildings in Exeter which are taller than 10m.'}
- Result: The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are commercial buildings, and the condition is that they must be taller than 10m. \n\n \noutput steps: ["Find Exeter 1 area", "Find commercial buildings in Exeter as many search results", "Apply condition for buildings taller than 10m"]

Tool: send_message

- Args: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

- Result: I found the area polygon for Exeter. The filtered results are saved in the artifact named `filtered_exeter_area_polygon`.

Tool: call_os_ngd

- Args: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}
- Result: Search results have been found. Artifact generated is : `exeter_area_polygon`, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..., count of records fetched : 98.

Tool: send_message

- Args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}
- Result: Filtered Exeter area polygons: 34 entries found. - Artifact name: `filtered_exeter_area_polygon` - Description: Polygon(s) from `exeter_area_polygon` specifically representing Exeter, filtered using `name1_text`, `description`, and `descriptiongroup`.

Tool: call_os_ngd

- Args: {'filters': ['Commercial Building'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': None, 'filename': 'commercial_buildings_exeter'}
- Result: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : `[buildings_commercial_buildings_exeter]`, Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Find commercial buildings in the area defined by the artifact 'filtered_exeter_area_polygon'. Filter for buildings that are commercial and taller than 10m. using filters ['Commercial Building'].", counts of records fetched are : [72895].

Tool: send_message

- Args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'buildings_commercial_buildings_exeter' to include only buildings where the 'relativeroofbase' (building height) is greater than 10 meters. Save the filtered results as 'commercial_buildings_exeter_over_10m'."}
- Result: Filtered commercial buildings in Exeter with relative roof base height > 10m. 200 out of 1814 buildings matched. - Artifact name: `commercial_buildings_exeter_over_10m` - Description: GeoDataFrame of commercial buildings in Exeter where the relative roof base height ('`height_relativeroofbase_m`') is greater than 10 meters. Includes all original columns and geometry.

Tool: send_message

- Args: {'target': 'plotting_agent', 'task_description': "Plot the artifact 'commercial_buildings_exeter_over_10m' to show commercial buildings in Exeter which are taller than 10m."}
- Result: Plotted 200 commercial buildings in Exeter over 10m in height. First 5 entries: ... (see above)

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons specifically representing Exeter
    # Use multiple columns for accuracy: name1_text, description, descriptiongroup
    exeter_mask = (
        (gdf['name1_text'].str.lower() == 'exeter') |
        (gdf['name1_text'].str.lower().str.contains('exeter')) |
        (gdf['description'].str.lower() == 'city') |
        (gdf['descriptiongroup'].str.lower().str.contains('settlement'))
    )
    filtered = gdf[exeter_mask]
    summary = f"Filtered Exeter area polygons: {len(filtered)} entries found."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'Polygon(s) from exeter_area_polygon specifically representing Exeter, filtered using name1_text, description, and descriptiongroup.'
    return [summary, artifact_name, artifact_description, filtered]
```

```
import matplotlib
matplotlib.use('Agg')
def filter_commercial_buildings_over_10m(data: list):
    gdf = data[0]
    # Filter for buildings where 'height_relativeroofbase_m' > 10
    filtered = gdf[gdf['height_relativeroofbase_m'] > 10]
    summary = f"Filtered commercial buildings in Exeter with relative roof base height > 10m. {len(filtered)} out of {len(gdf)} buildings matched."
    artifact_name = 'commercial_buildings_exeter_over_10m'
    artifact_description = "GeoDataFrame of commercial buildings in Exeter where the relative roof base height ('height_relativeroofbase_m') is greater than 10"
    return [summary, artifact_name, artifact_description, filtered]
```

```
import matplotlib
matplotlib.use('Agg')
def plot_commercial_buildings_exeter_over_10m(data: list):
    import geopandas as gpd
    import folium
```

```
import pandas as pd
import os

gdf = data[0].copy()
# Convert all columns with Timestamp to string for folium compatibility
for col in gdf.columns:
    if pd.api.types.is_datetime64_any_dtype(gdf[col]):
        gdf[col] = gdf[col].astype(str)
# Ensure CRS is EPSG:4326 for folium
if gdf.crs is not None and gdf.crs.to_epsg() != 4326:
    gdf = gdf.to_crs(epsg=4326)

# Create folium map centered on Exeter (approximate center from data)
centroid = gdf.geometry.centroid.unary_union.centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13, tiles='cartodbpositron')

# Add buildings to map
folium.GeoJson(
    gdf,
    name='Commercial Buildings >10m',
    tooltip=folium.GeoJsonTooltip(fields=['osid', 'buildinguse', 'height_relativeroofbase_m', 'height_absoluteroofbase_m'],
                                   aliases=['ID', 'Use', 'Height (relative, m)', 'Height (absolute roof base, m)'])
).add_to(m)

# Save map
map_filename = 'commercial_buildings_exeter_over_10m_map.html'
m.save(map_filename)

# Prepare summary
summary_rows = gdf[['osid', 'buildinguse', 'height_relativeroofbase_m', 'height_absoluteroofbase_m']].head(5)
summary = f"Plotted {len(gdf)} commercial buildings in Exeter over 10m in height. First 5 entries:\n" + summary_rows.to_string(index=False)
description = "Folium map showing all commercial buildings in Exeter with a relative roof base height greater than 10m. Each building is interactive with t
return [summary, map_filename, description, None]
```